NGUYEN ANH TUAN

Full-Stack Engineer

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ABOUT

Computer science and security engineer. Since 2025, full-stack TypeScript/Node.js developer on the central console of a zero-trust network security (ZTNA) platform, in a two-developer team working directly with the CTO, taking features from spec to production.

PROFESSIONAL EXPERIENCE

Full-Stack Engineer (fixed-term contract) | Snowpack (ZTNA platform) 05/2026 - Present
Paris

- **Node.js upgrade (18 to 24)**: pre-compiled the backend. Start-up 2.5 to 1.5 min, every CI pipeline 1-1.5 min faster, deployed image 30% smaller, about 40 hidden type errors caught by the new build check.
- **Production incident**: CPU spiking every 10 minutes, traced in Grafana to background jobs piling up and one client device's ever-larger reports. Fixed both (jobs skip a run while the last is still going; 2 MB limit from 3 months of real data). CPU from 100% at peak to 10-20% max.
- **Partner app identity (Android SDK)**: designed how our app verifies which partner app is calling, using the identity the phone's OS checks; the allowed list works offline.
- **Self-hostable console**: packaged for clients' own offline servers as one executable plus its dependencies (Node SEA), on standard and ARM machines (Raspberry Pi).
- **Audit log**: records sign-ins and the actions taken in each company's space.
- **Stack**: TypeScript, Node.js, ESM, swc, esbuild, Node SEA, k6, Grafana, Docker, Vitest, GitLab CI.

Full-Stack Developer (internship) | Snowpack (ZTNA platform) 02/2025 - 03/2026
Paris

- **Devices, 7 weeks after joining**: IoT devices as a new kind of account that signs in with a token instead of a password, with its own access control, backend and frontend.
- **Group-based access, shipped in under 2 months**: from the CTO's idea, wrote the spec and built it: bulk assignment of users to services, inherited access, change log, 26 tests. Later extended to licences and devices.
- **Identity-provider sync** (Keycloak, Microsoft Entra ID): each company's groups copied every 5 minutes and at sign-in; closed a hole letting our server be pointed at internal addresses.
- **Typed task queue**: replaced a failing lock and fixed a double-licence bug; misuse caught at compile time. Now 12 queues, reused by another developer without help.
- **Faster data loading on every console page** (users page 10-15 s down to 4-5 s, 2-3x faster): replaced a shared "reload everything" with per-page loading and a short cache.
- **Data usage tracking**: daily usage per user, device and partner, built from the raw usage logs, with the admin usage charts and a migration that rebuilt the history.
- **Stack**: TypeScript, Node.js, TSOA / OpenAPI, React, MongoDB, TypeORM, Keycloak, Docker, GitLab CI.

PROJECTS

RFP proposal scorer | 24h hackathon - architecture lead, team of 4 09/2026
Frankfurt

- **AI reviewer for sales proposals**: rules in code for amounts and dates; the AI model (Gemini Flash) scores 7 criteria and must quote the proposal; code checks each quote exists.
- **Tested on 10 real public bids** (State of Michigan): matched the State's ranking on 2 of 3 contracts, under $0.60 per bid.
- **Stack**: Python (FastAPI), React · github.com/UNtergas/SiviHack-2026-Team

EDUCATION

INSA Centre Val de Loire | Engineering degree, Security and Information Technology 2020 - 2026
Bourges (2022-2026) · Selective Franco-Vietnamese integrated preparatory programme, Hué (2020-2022)

SKILLS

Languages: TypeScript, JavaScript, Python, SQL, Java (academic)
Backend: Node.js, Express, TSOA, REST, OpenAPI / Swagger, FastAPI
Frontend: React, Next.js, Vite
Databases: MongoDB, TypeORM, SQL
Security & Identity: Authentication, RBAC, application security, Keycloak, Microsoft Entra ID
Testing: Vitest (TDD, end-to-end), k6 (load testing)
Build & DevOps: Docker, GitLab CI, ESM, swc, esbuild, Node SEA, Grafana
Practices: OOP, TDD

LANGUAGES

English French Vietnamese